Question 1

Two channels, one with a bit rate of 190 kbps and another with a bit rate of 180 kbps, are to be multiplexed using pulse-stuffing TDM with no synchronization bits. Answer the following questions: (12 marks)

a. What is the size of the frame?

b. What is the frame rate?

c. What is the duration of the frame?

d. What is the data rate of the link?

Question 2

Draw the graph of the signal for the bit stream of 1(five consecutive) then a zero, then 1(five consecutive) knowing that the most recent non-zero signal level for bit 1 was negative for AMI, for pseudoternary most recent bit 0 was negative : (9 marks)

a. NRZ-L

b. Bipolar AMI

c. Psuedoternary

Question 3

A voice signal with a frequency range from 300 Hz to 3000 Hz is sampled at a rate of 7000 samples per second.

a. If SNR=30dB, find the number of quantization levels (L).

b. What is the recommended data rate?

Question 4

A light signal is traveling through a fiber-optic cable. What is the propagation delay in the signal if the length of the cable is 10 m, 100 m, and 1 km (assume a propagation speed of 2 10^8 m)?

Question 5

How many bits per baud can be sent in each of the following cases if the signal constellation has the given number of points? (11 marks)

a. 2 b. 4 c. 16 d. 1024